INCH-POUND

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SUPERSEDING
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24 January 1986

MILITARY SPECIFICATION

SWEATSHIRT, HOODED, PHYSICAL FITNESS UNIFORM (PFU)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a knitted jersey fleece sweatshirt used as a component of the Physical Fitness Uniform.

1.2 Classification. The sweatshirt shall be of one type in the following sizes (see 6.2).

Sizes

X-small		Large
Small	Medium	X-large

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

V-F-106	-	Fasteners, Slide, Interlocking
DDD-L-20	-	Label: For Clothing, Equipage and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-E-20652	-	Eyelet, Metallic, and Eyelet Washers, Metallic
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment, Equipage: General Specification For
MIL-T-43548	-	Thread, Polyester Core: Cotton-, Rayon-, or Polyester-Covered
MIL-T-43624	-	Thread, Polyester, Spun

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members For Containers and Pallets
MIL-STD-1491	-	Glossary of Knitting Imperfections

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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FEDERAL TRADE COMMISSION

Rules and Regulations Under the Flammable Fabrics Act

(Copies are available from the Federal Trade Commission, Washington, DC 20580.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

Dimensional Changes in Automatic Home Laundering of Garments

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215,Research Triangle Park, NC 27709.)

COLOR ASSOCIATION OF THE UNITED STATES, INC.

Department of Defense Standard Thread Shades

(Application for copies should be addressed to the Color Association of the United States, Inc., 343 Lexington Ave., New York, NY 10016. If color samples are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Samples

3.2.1 Guide samples. End item samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.2.2 Standard sample. The finished cloth shall match the standard sample (see 6.4) for shade and appearance. Unless otherwise indicated, the finished cloth shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 3.3.5.1.1).

3.3 Material. The cloth shall be such that the end item will conform to the Rules and Regulations of the Flammable Fabrics Act without the application of a flame retardant finish to the material or to the end item. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Jersey knit fleece cloth. The cloth for the body, hood, sleeves and muff pockets shall be a jersey knit fleece cloth consisting of 58 ± 5 percent cotton, 1 to 2 percent rayon, and the remaining percentage acrylic fiber. The cloth shall be knit using the following facing, tie-in, and backing yarns (see 6.5.1):

Facing yarn - Blend of a minimum of 96 percent carded cotton with the remaining percentage rayon
Tie-in yarn - Unrelaxed high bulk acrylic or any acrylic yarn that will meet all specified requirements
Backing yarn - Blend of a minimum of 80 percent carded cotton with the remaining percentage acrylic fiber.

The cloth shall be napped on the back such that the degree of nap and appearance of nape approximate that of the standard sample. In the event that there is no standard sample, there shall be a "medium" degree of nap of sufficient density to prevent show through of backing and/or tie-in yarns. The finished cloth shall conform to the following requirements when tested as specified in 4.4.1.2.

Weight, ounces per square yard	8.2 ± 0.5
Wales per inch	21 (minimum)
Courses per inch	25 (minimum)
Bursting strength, pounds	75 (minimum)

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3.3.2 Rib knit cloth. The cloth for cuffs, waistband, and side panels shall be a plaited 1 x 1 rib knit cloth consisting of 47 ± 5 percent cotton, 2 to 3 percent rayon, and the remaining percentage acrylic fiber. The cloth shall be knit using the following facing and inside yarns (see 6.5.2):

Facing yarn - Blend of a minimum of 96 percent carded cotton with the remaining percentage rayon
Inside yarn - Unrelaxed high bulk acrylic or any acrylic yarn that will meet all specified requirements

The finished cloth shall conform to the following requirements when tested as specified in 4.4.1.2.

Weight, ounces per square yard	6.5 ± 0.5	
Wales per inch	18	(minimum)
Courses per inch	25	(minimum)
Bursting strength, pounds	90	(minimum)

3.3.3 Finish. No flame retardant finish or flame resistant finish shall be applied to the cloth or to the end item.

3.3.4 Color. All cloth for the sweatshirt shall be Light Heather Gray 446.

3.3.4.1 Colorfastness. The finished cloth shall show fastness to light, laundering (after three cycles), and perspiration equal to or better than the standard sample, or equal to or better than the rating of "good" when tested as specified in 4.4.1.2. The finished cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating for crocking not lower than 3.5 when tested as specified in 4.4.1.2.

3.3.4.2 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and having a correlated color temperature of 7500 ± 200K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K.

3.3.5 pH. The pH of the finished cloth shall be 5.0 to 8.5 when tested as specified in 4.4.1.2.

3.3.6 Thread. The thread for seaming and stitching shall be cotton-covered polyester conforming to MIL-T-43548 or spun polyester conforming to MIL-T-43624, dyed light gray DoD Standard Thread Shade AX, CA 66064. The thread shall show colorfastness to laundering, perspiration, and light equal to or better than the standard sample or equal to or better than the rating of "good".

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3.3.7 Slide fastener. The slide fastener shall be continuous element or individual element molded plastic, style 8, size MS, type IV, with long pull tab, conforming to V-F-106, except that the tape width shall be 9/16 ± 1/32 inch wide. The color, including chain, tape, and slider shall match the basic fabric.

3.3.8 Drawcord. The drawcord shall be 3/16 inch diameter 100 percent filament nylon, hot-knifed into lengths of 45 to 46 inches. The nylon shall be bright, regular or high tenacity polymide having a 840 nominal denier and having a minimum breaking strength of 150 pounds. The drawcord shall be a basket weave with no core. The drawcord shall be dyed light gray DoD Standard Thread Shade sample AX, C.A. 66064 and shall show "good" colorfastness to laundering.

3.3.9 Eyelet. The eyelets for the hood drawcord shall be one piece, brass (BNE133), conforming to MIL-E-2064?/1. As an option, a washer equal to or better than Stimpson Part No. C-84 may be used with the eyelet (see 6.6).

3.3.10 Ink. The ink for screen printing "ARMY" to the sweatshirt shall be equal to Ruthland Plastic, Inc., product, HO #394, black thermal plastic base ink (withstanding temperatures to 300°F) or equal to all the requirements cited.

3.3.11 Label. Each sweatshirt shall have a combination size, identification, and instruction label conforming to type VI, class 14 of DDD-L-20. All required information shall be printed on one side, or the size and identification information may be on one side and care instructions on the other side provided the care instructions are right side up when the label is turned over. All requirements including size of characters shall be as specified in DDD-L-20. The label shall show fastness to laundering. The body of the instruction label shall be as follows:

CARE INSTRUCTIONS

Hand or machine wash in cold or warm water (maximum 120°F).
Do not Bleach or Iron. Tumble dry at low setting (maximum 140F).

3.4 Design. The sweatshirt shall be hooded, long sleeve, and have a front slide fastener closure. The hood shall be double thickness with eyelets and drawcord, the body includes 3-1/2 inch wide double thickness rib knit side panels and split muff pockets. The cuffs and waist shall be rib knit finished.

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3.5 Construction. The construction shall conform in all respects to the requirements specified in table I and herein. Figures 1 and 2 are furnished solely for guidance and information. Should variation from this specification appear in figure 1, this specification shall govern.

3.5.1 Stitches, seams, and stitching. All stitches, seams, and stitching shall conform to FED-STD-751. The type of seam, stitching, and stitches per inch shall be as specified in table I. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used.

3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/2 inch except where ends are turned under or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewn.

3.5.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during end item inspection shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.5.1.2 Types 504, 506, and 607 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.5.1.1.1a. and 3.5.1.1.1b. The gage of the overedge stitching shall be 3/16 ± 1/16 inch. The guide and knives for the overedge machine shall be set to trim only the ravelled edges of the fabric.

3.5.2 Marking. The black "ARMY" marking shall be screen printed on the front of the sweatshirt as shown in figure 1. Letters shall measure 5/8 ± 1/16 inch in width and 3 ± 1/4 inches in height (see figure 2). The overall width shall be 12 ± 1/4 inches and shall be positioned as specified below. Printing shall show good colorfastness after 15 launderings when tested as specified in 4.4.1.1.

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Size	Marking location Distance down from neck seam ± 1/2 inch)
X-Small	3
Small	3-1/2
Medium	4
Large	4-1/2
X-Large	5

3.5.3 Thread breaks and ends of seams. Ends of all seams and stitching when not caught in other seams shall be backtacked not less than 1/4 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks shall be secured by stitching back of the break not less than 1/2 inch.

3.6 Preshrinking. The knitted cloth shall be preshrunk by a mechanical shrinkage process before being cut, such that the sweatshirt will meet all shrinkage requirements specified in 3.9.

3.7 Manufacturing operations requirements. The sweatshirt shall be manufactured in accordance with all operations specified in table I. The contractor is not required to follow the exact sequence of operations as long as all requirements are met.

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NO.	TABLE I MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. The sweatshirt shall be cut so that the body and cuff parts of the fleece shall be cut widthwise (reverse weave) in the direction of the wales. The sleeves and hood shall be cut lengthwise in the direction of the wales. b. Cut all the component parts of the fleece cloth for the sweatshirt from one piece of fabric. c. Cut the side panels, waistband, and cuffs from the same piece of rib knit cloth. The wales of the rib knit shall be in the same direction as the sleeve and hood fleece. The rib knit shall be of sufficient width to allow for double thickness and finished width.						
2.	Replacement of defective components. Care shall be exercised during the spreading, cutting and manufacturing operations to assure that material defects and damages as specified in 4.4.2 are excluded, and replaced with non-defective and properly matched material.						

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TABLE I (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Shade and size marking. Mark, ticket or bundle all component parts to insure a uniform shade, size and proper assembly throughout the sweatshirt. Any method of marking may be used except: a. Corrosive metal fastening devices. b. Sew on tickets. c. Adhesive type tickets which leave an impression or traces of paper or adhesive on the material after removal of the ticket.						
4.	Assemble hood. a. Fold hood in half lengthwise, fold in half lengthwise again and seam down back open edge. b. Open hood at corners and attach eyelets 1-1/2 + 1/4 -1/2 inch up from the neck seam allowance and 3/4 ± 1/4 inch in from fold on both sides. The eyelets shall not differ more than 1/4 inch in either measured direction. c. Insert cord through eyelets and double knot both ends. Turn hood right side out over drawcord.	504	SSa-1	10-12	50/2	50/2	50/2
5.	Seam cuffs. a. Fold in half widthwise and seam. b. Fold in half lengthwise with right sides out.	504	SSa-1	10-12	50/2	50/2	50/2

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NO.	TABLE I (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Attach muff pockets.						
	a. Turn edge of opening to inside 7/8 ± 1/8 inch and stitch along raw edge the entire length of opening.	406	EFa-1	8-12	50/2	50/2	50/2
	b. Align fronts and bottoms of sweatshirt to edges of muff pockets (right and left respectively) and turning the top and side of pockets under 1/2 ± 1/8 inch stitch in place, making sure pocket opening is flat and smooth.	301	LSb-1	10-12	50/2	50/2	
	c. Bartack both sides of opening, positioned over one line stitch.	Bartack		28 per Bartack	50/2	50/2	
7.	Join shoulder.						
	a. With right sides together stitch front to backs at the shoulders.	504	SSa-1	10-12	50/2	50/2	50/2
	b. With seam flat, cover seam shoulders.	406	SSh-2	8-12	50/2	50/2	50/2
8.	Assemble sweatshirt.						
	a. With right sides together stitch fronts to side rib knit panels (double thickness) and panels to back piece.	504	SSa-1	10-12	50/2	50/2	50/2
	b. Position hood with ends 1/2 ± 1/8 inch in from both ends of neckline and stitch in place catching labels at center back, with the size visible.	504	SSa-1	10-12	50/2	50/2	50/2

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TABLE I (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Assemble sweatshirt. (cont'd)						
	c. With right sides together close sleeve seams, then position to armhole matching sleeve seam to center of side insert gusset (half mark on sleeve to match shoulder seam).	504	SSa-1	10-12	50/2	50/2	50/2
	d. With right sides together and seams aligned, stitch cuffs to sleeves.	504	SSa-1	10-12	50/2	50/2	50/2
	e. Fold waist ribbing in half lengthwise with right sides out and stitch to the bottom of the sweatshirt.	504	SSa-1	10-12	50/2	50/2	50/2
	f. Coverseam all seams (except muff pocket tops and sides).	406	SSh-2	8-12	50/2	50/2	50/2
	g. Overedge stitch both sides of front opening.	504	EFd-1	10-12	50/2	50/2	50/2
	h. Fold front under 1/2 ± 1/16 inch position slide fastener with teeth to edge and stitch in place turning top under. The bottom fastener shall be more no than 1/2 inch from the bottom and 3/8 ± 1/8 inch from the hood seam.	301	LSb-1	10-12	50/2	50/2	
	or						

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NO.	TABLE I (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Assemble sweatshirt. (cont'd)						
	as an alternate to G and H						
	g. (1) Stitch slide fastener tape to edge of front opening.	504	LSbu-2(a)	10-12	50/2	50/2	
	h. (1) With slide fastener attached, fold front under 5/8 ± 1/16 inch with teeth to edge of fold and stitch in place turning top under. The bottom of the fastener shall be no more than 1/2 inch from the waistband bottom and the top of the fastener shall be 3/8 ± 1/8 inch from the hood seam.	301	LSbu-2(b)	10-12	50/2	50/2	
	i. Adjust drawcord for even lengths on each end and tack top center edge of hood drawcord.	Bartack		28 per bartack	50/2	50/2	
9.	Clean sweatshirt.						
	Trim and remove all threads inside and outside of sweatshirt.						
10.	Lettering.						
	Screen print "ARMY" on the front of the sweatshirt.						

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3.8 Finished measurements. The measurements of the finished sweatshirt shall conform to the requirements specified in table II. All measurements shall be taken without stretching the material.

TABLE II. Finished measurements (in inches)

		Extra Small	Small	Medium	Large	Extra Large	Tolerance
1/2 Chest width	A/	19 1/2	21 1/2	23 1/2	25 1/2	27 1/2	+ 1 - 1/2
Length	B/	24	25	26	27	28	± 1
Armhole	C/	9-1/2	10	10-1/2	11	11-1/2	± 1/2
Sleeve overarm	D/	24-1/4	25-1/4	26-1/4	27-1/4	28-1/4	± 3/4
Sleeve underarm	E/	20 3/4	21 3/4	22 3/4	23 1/4	23 3/4	± 3/4
Sleeve opening	F/	5-1/2	5-1/2	6	6	6-1/2	± 1/2
Neck opening relaxed	G/	10	10-1/2	10-1/2	11	11	± 1/2
Sleeve trim	H/	3	3	3	3	3	± 1/4
Bottom trim	I/	3	3	3	3	3	± 1/4
Side panel width	J/	3-1/2	3-1/2	3-1/2	3-1/2	3-1/2	± 1/4
Hood width	K/	10-1/2	11	11	11-1/2	11-1/2	± 1/2
Hood length	L/	13 1/2	14	14	14 1/2	14 1/2	± 1/2
Pocket width	M/	6-3/4	6-3/4	6-3/4	6-3/4	6-3/4	± 1/2
Pocket length	N/	8-3/4	8-3/4	8-3/4	8-3/4	8-3/4	± 1/2

A/ Width - just below armhole, flat across the chest as shown on figure 1.

B/ Length - high point of shoulder to bottom including the rib.

C/ Armhole - shoulder seam point to underarm straight point to point.

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D/ Sleeve overarm - shoulder seam to sleeve including rib cuff.

E/ Sleeve underarm - underarm seam to sleeve end including rib cuff.

F/ Sleeve opening - measure at cuff seam with rib extended until the body fabric lays flat.

G/ Neck opening relaxed - fold center front to center back, measure seam.

H/ Sleeve trim - measure from seam to end of double thickness rib cloth.

I/ Bottom trim - measure from seam to end of double thickness rib cloth.

J/ Side panel width - measure seam to seam of double thickness of cloth.

K/ Hood width - lay flat, measure front fold to back seam.

L/ Hood length - lay flat, measure top fold to neck seam in center of hood.

M/ Pocket width - center front fold at slide fastener to edge of pocket at widest point.

N/ Pocket length - top edge of pocket to rib bottom seam.

3.9 Dimensional stability. The finished sweatshirt after five launderings and dryings shall not elongate more than 5.5 percent or shrink more than 8.0 percent for the lot average, with the average of no sample unit to exceed 10.0 percent for the sleeve length, body length and chest width, when tested as specified in 4.4.4.

3.10 Workmanship. The finished sweatshirts shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3, and tested as specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the characteristics listed below conform to the specified requirements.

Characteristic	Requirement paragraph
Cloth flammability	3.3
Yarns, fiber identification and content	3.3.1 and 3.3.2
Cloth fiber content	3.3.1 and 3.3.2
Drawcord, material identification, denier and breaking strength	3.3.8
Ink, material identification	3.3.10

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4.2.1.2 Component testing. The jersey knit fleece and rib knit cloths shall be tested for the characteristics listed in table III. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table III shall be followed. The sample unit shall be 1 yard full width. The lot shall be rejected if one or more sample units fail to meet any requirement specified. The sample size shall be in accordancew with the following:

Lot size (yards)	Sample size (sample units)
Up to 22,000	2
22,001 and over	3

TABLE III. Cloth component tests

Characteristic	Requirement reference	Test method
Weight (oz./sq. yd.)	3.3.1 and 3.3.2	5041
Wales and courses per inch	3.3.1 and 3.3.2	5070
Bursting strength	3.3.1 and 3.3.2	5120
Colorfastness to:		
Light (after 80 standard fading hours	3.3.4.1	5660
Laundering (after 3 cycles	3.3.4.1	5614 1/
Perspiration	3.3.4.1	5680 1/
Crocking	3.3.4.1	5651 1/
pH	3.3.5	2811
Screen printing on jersey knit, colorfastness to laundering	3.5.7	5614 2/

1/ If material shade is obtained by a percent black solution-dyed rayon fiber mixed with 96 percent natural face cotton in the facing yarns, a certificate of compliance shall be submitted and will be accepted for the stated requirement.

2/ Use fifteen laundering cycles with drying after each cycle.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table IV. The lot size shall be expressed in units of sweatshirts. The sample units shall be one sweatshirt. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE IV. End item visual defects

Examine	Defect	Classification Major	Classification Minor
Material defects	Any hole, cut, tear, dropped stitches or run, uneven napping, or mend 1/	101	
	Uneven, ridgy or cockled cloth 1/		201
	Fleeced cloth not napped on inside	102	
	Any slub (more than three times normal diameter of yarn and clearly visible) on outside of sweatshirt 1/		202
	Snag 1/		203
	Permanent crease in fabric	103	
Color and shade	Color not as specified	104	
	Variation in shade within a part or between two or more parts		204
	Streaks clearly visible		205
Cleanness	Spot or stain, clearly noticeable:		
	On outside:		
	- more than 1/4 inch	105	
	- up to 1/4 inch inclusive		206
	On inside		207
	Rancid odor	106	
	Any marking clearly visible on outside	107	
	One or more marking tickets not removed		208
	Thread ends not trimmed, or loose threads not removed from sweatshirt		209
Label	Omitted, incorrect, or illegible, misplaced, or not attached as specified	108	
Waistband and cuffs	Not smooth, has pleats, is twisted or is stretched	109	
	Not specified width	110	
Sleeves	Fullness not distributed evenly	111	
	Length not as specified	112	
Slide fastener	Puckers or pleats around fastener	113	
	Fastener stops more than 1/2 inch from neck seam or from bottom of sweatshirt	114	
	Fastener not working properly	115	

1/ For knitted fabric imperfection definitions, see MIL-STD-1491.

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Pockets	Misplaced or unevenly matched by more than 1/4 inch	116	
Eyelets	One or more omitted	117	
	Not tightly clinched	118	
	Out of alignment by 1/4 inch in either direction		210
Drawcord	Not tacked or double knotted		211
	Ends not hot knifed		212
Lettering	Omitted, incorrect, or illegible	119	
	Misplaced or not applied as specified		213
Component and assembly	Any component part or required operation omitted or added (unless otherwise classified herein	120	
	Any operation not performed as specified (unless otherwise classified herein)		214
	Cuffs, side panels and waistbands not double thickness of rib knit cloth	121	
	Any component part not as specified	122	
Cutting	Any component part not cut in accordance with directional lines as indicated on pattern or in accordance with specification requirements	123	
Seams and stitching	Any open seam (except on hems)	124	
	Raw edges:		
	- any on outside	125	
	- on inside, 1/8 to 1/2 inch (inclusive)		215
	- on inside, over 1/2 inch	126	
	Runoff, on any line of stitching not catching all plies of material to be seamed:		
	- up to 1/2 inch (inclusive)		216
	- more than 1/2 inch	127	
	Stitch tension:		
	- loose tension resulting in a loose seam	128	
	- tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	129	

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Seams and stitching (cont'd)	Up to two stitches per inch less than specified minimum		217
	Three or more stitches per inch less than specified minimum or more than specified maximum	130	
	Bartack not as specified	131	
	Open seam on hems:		
	- 1/4 to 1 inch		218
	- over 1 inch	132	
NOTE:	A seam shall be considered as being open when one or more stitches joining a seam are broken, or when two or more successive skipped stitches occur.		
	Seam or stitch type not as specified	133	
	Parts of sweat shirt badly pleated, caught or twisted in any type of stitching	134	
	Seam twisted or puckered		219
	Thread breaks (all stitch types) backstitched less than 1/2 inch		220
	Ends of all seams and stitchings produced with 301 stitch type not caught in other seams or stitchings, or backtacked less than 1/2 inch		221
	Ends of a continuous line of stitching overlapping less than 1/2 inch		222

4.4.3 End item dimensional examination. The end items shall be examined for conformance to the measurements specified in table II. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of sweatshirts. The sample unit shall be one sweatshirt. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Dimensional stability test. The end items shall be tested for shrinkage as specified in Method 105-1987 of AATCC using normal machine cycle 120° ± 5°F washing temperature, and delicate dryer setting (140°F).

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The sweatshirts shall not be moistened or pressed before measuring. Measurements shall be taken to the nearest 0.1 inch with the sweatshirts laid flat. Elongation shall be reported with a minus ("-") sign. The lot size shall be expressed in sweatshirts. The sample unit shall be three sweatshirts from each size in the lot. The sample size shall be as indicated in 4.4.1. The lot shall be rejected if the elongation or shrinkage requirements specified in 3.9 are not met.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container. Open and non-continuous heat sealed seams and closure of polyethylene bag, omission of vent hole in polyethylene bag.
Content	Number of unit packs per shipping container is more or less than specified.

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each sweatshirt shall be neatly folded to approximate 15 by 12 inches and inserted in a close-fitting clear polyethylene bag. The bag shall be formed with heat sealed seams that are straight, continuous, and parallel to each other and the formed edges of the bag. The bag may be fabricated from polyethylene film tube or sheeting. The final closure of the bag may be heat sealed or a reverse flap or tuck. A 1/4 inch diameter hole shall be made at one corner of the bag to allow air to escape.

5.1.2 Commercial. The sweatshirt shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Twelve sweatshirts of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The inside dimensions of each shipping container shall approximate 24-1/2 inches in length, 15-1/2 inches in width, and 11 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced in accordance with the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to non-metallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Twelve sweatshirts of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, minimum grade 200 of PPP-B-636. The inside dimensions of each shipping container shall approximate 24-1/2 inches in length, 15-1/2 inches in width, and 11 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

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5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping containers shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III, as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Sweatshirts of one size only, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), sweatshirts packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 3 in accordance with appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special markings required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The label shall indicate the following information:

Stock number
Nomenclature
Size
Quality

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The sweatshirts are intended for use as a component of the Physical Fitness Uniform.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.2, 4.3, and 6.3).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to end item guide samples and/or standard samples, address the contracting activity issuing the invitation for bids.

6.5 Knitting cloth.

6.5.1 Jersey knit fleece cloth. A fabric meeting the requirements of this specification was knitted with the following yarns:

Facing: 28/1 cotton count-96/4 carded cotton/solution dyed black rayon.

Tie-In: 1/40 worsted count-unrelaxed Orlon, Type 42.

Backing: 14/1 cotton count-80/20 cotton/acrylic.

6.5.2 Rib knit cloth. A fabric meeting the requirements of this specification was knitted with the following yarns:

Facing yarn: 28/1 cotton count-96/4 carded cotton/solution dyed black rayon.
Inside yarn: 1/40 worsted count-unrelaxed Orlon, Type 42.

6.6 Source of supply. A source of supply for the eyelets is:

Stimpson Co., Inc.
900 Sylvan AVenue
Bayport, NY 11705-1097

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6.7 Subject term (key word) listing.

Athletic clothing
Knitted cloth

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - GL

Review activities:
Army - MD
DLA - CT

Preparing activity:
Army - GL

(Project 8415-A688)

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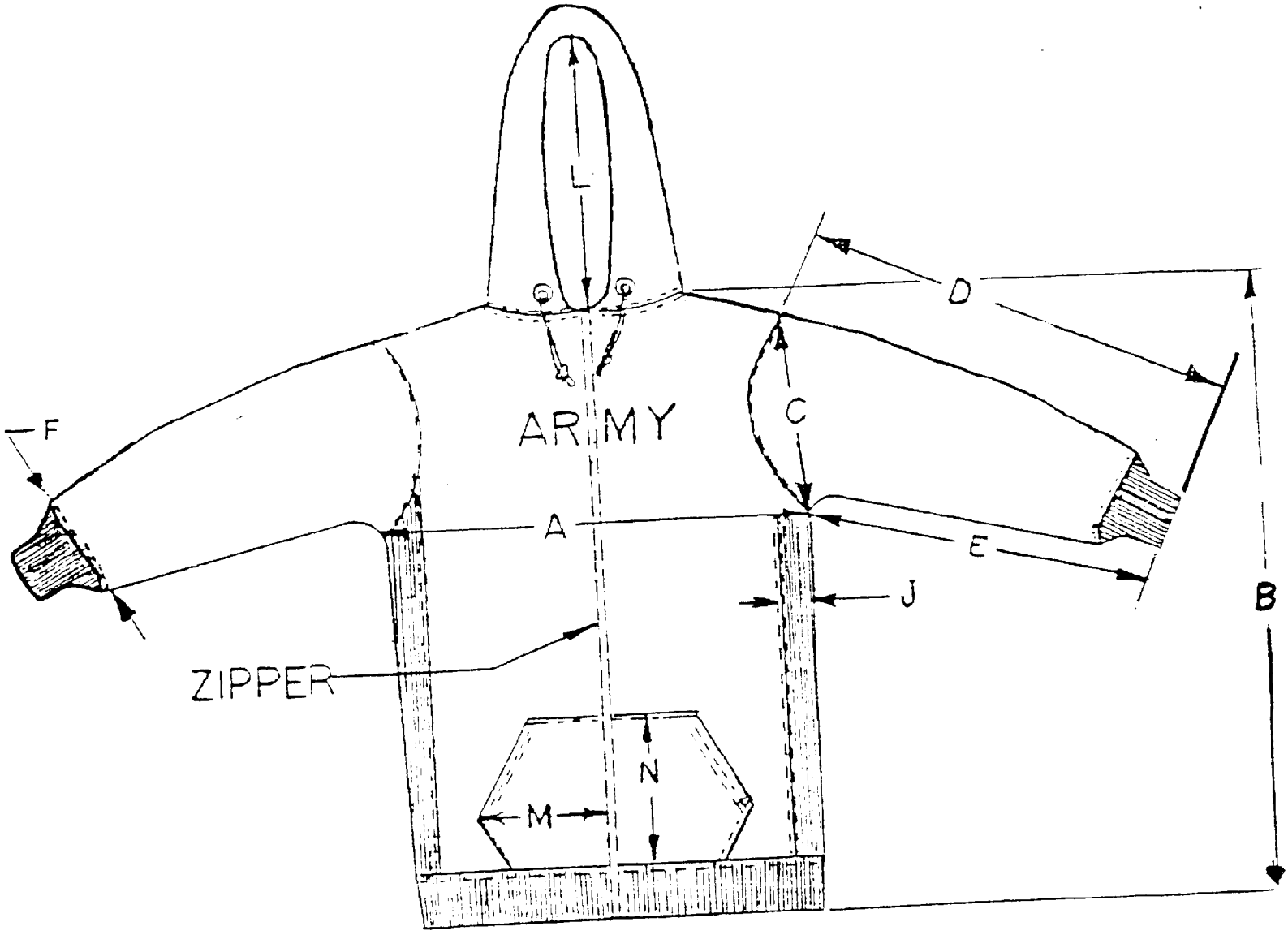


FIG.1 SWEATSHIRT
PHYSICAL FITNESS UNIFORM

2-1-1631

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NOTE: The width of each letter shall be 5/8 ± 1/16 inch. The style of lettering shall be as shown above.

FIGURE 2

MIL-C-44436A(AS)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-44212A(GL)	Sweatshirt, Hooded, Physical Fitness Uniform (PFU)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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